

**Basic interactive script `x4pro.sh` for running demo-codes.
TUI - Text-based User Interface.**

```
# Script: x4pro.sh
# Author: Viktor Zerkhin
# Created: 2026-02-13
# Version: 2026-02-23
# License: MIT
# Run: $ [bash] x4pro.sh [{options|files}]
# --help display this text
# Example: $ bash x4pro.sh
# $ ./x4pro.sh
# $ bash x4pro.sh --help
# Commands:
# A - run all tests and exit
# C - clean log file, remove generated output files
# # - run example number #, where #:[0..26]
# #-# - run examples by range of numbers, e.g. 1-5
# L - list examples
# S - statistics of the local EXFOR-database
# V - versions of system components
# O - exit
```

1) Start: welcome, versions

```
C:\WINDOWS\system32\cmd.exe - bash x4pro1.sh
G:\projects\zerkin\pdev\x4pro7-dist\x4pro14>bash x4pro1.sh

X4Pro
-----
X4Pro: fully relational EXFOR database.
      Professional Edition.
      Version 2026-02-23.
      Demo-examples on Python/SQL/Fortran:
      search/retrieve/plot/recalculate data.
      v.zerkin@gmail.com, IAEA-NRDC, 2021-2026
-----
Run:      2026-02-24,13:04:27
Platform: MINGW32_NT-6.2 1.0.19(0.48/3/2) i686
Computer:  ZVV-MINI-PC
Shell:     GNU bash, version 3.1.23(1)-release (i686-pc-msys)
Bash-ver:  3.1.23(1)-release
Script:    x4pro1.sh
Now Dir:   /g/projects/zerkin/pdev/x4pro7-dist/x4pro14
---Versions---
1) python: [3.9.5]
2) pip:    [21.1.2]
3) matplotlib: [3.5.2]
4) plotly:  [5.3.1]
5) requests: [2.26.0]
6) pandas:   [1.3.3]
7) kaleido: [0.2.1]
8) couchdb: [1.2]
9) gfortran: GNU Fortran (MinGW.org GCC Build-2) 9.2.0
10) sqlite3: 3.47.2 2024-12-07 20:39:59 (64-bit)
11) dialog:   Version: 1.1-20100119
12) database-file: ../x4sqlite1.db date:2026-02-16 size=186M
    database-data: 2026-02-16,12:31:45 Entry:390 Datapoints:251401

x4pro>
```

2) [S] - EXFOR database statistics

```

C:\WINDOWS\system32\cmd.exe - bash x4pro.sh
x4pro> s
---EXFOR Statistics---
Database-file: ../x4sqlite1.db 2026-02-16,14:52:50 size=186M
Data in database:
--Database-uploaded: 2026-02-16,12:31:45
--Data-update: 2026-01-22
--Entry/Subent: 390/6692
--Datasets/points: 5380/251401
--wget-cache: 81/2026-02-16,14:52:49
--Datasets by Targets: (limit:8)
+-----+
| Datasets | Target |
+-----+
| 285      | O-16   |
| 233      | Al-27  |
| 162      | U-235  |
| 145      | Mn-55  |
| 127      | Co-59  |
| 122      | Ni-58  |
| 110      | C-12   |
| 109      | Fe-56  |
+-----+
--Datasets by Incident particles: (limit:8)
+-----+
| Datasets | Projectile | Help      |
+-----+
| 4846     | N          | Neutrons  |
| 540      | P          | Protons   |
| 32       | HE3        | He-3      |
| 22       | A          | Alphas    |
| 17       | D          | Deuterons |
| 9        | He-6       |           |
| 4        | 0          | -         |
+-----+
--Datasets by Quantity: (limit:8)
+-----+
| Datasets | Quantity | Help      |
+-----+
| 3765     | CS       | Cross section data |
| 503      | DA       | Differential data with respect to angle |
| 251      | DAP      | Partial differential data with respect to angle |
| 170      | FY       | Fission product yields |
| 156      | RI       | Resonance integrals |
| 136      | DAE      | Differential data with respect to angle and energy |
| 133      | CSP      | Partial cross section data |
| 127      | POL      | Polarization data |
+-----+
x4pro>

```

3) [L] - List of demo-folders

C:\WINDOWS\system32\cmd.exe - bash x4pro.sh

```
x4pro> 1
---EXAMPLES---
  part1:: Retrieve and plot EXFOR data
0  part1-0-sig      Retrieve and plot EXFOR cross sections (plain Python-program)
1  part1-1-sig      Retrieve and plot EXFOR cross sections (modular Python-program)
2  part1-2-da       EXFOR angular distributions
3  part1-3-dap      EXFOR partial angular distributions
4  part1-4-de       EXFOR energy spectra of outgoing particles
5  part1-5-dae      EXFOR double differential cross sections
6  part1-6-fy       EXFOR fission yield
7  part1-7-covar    Retrieve and plot covariance data stored in EXFOR
8  part1-fortran     Retrieve data from EXFOR using SQL from Fortran programs
  part2:: Retrieve and plot EXFOR data and ENDF data from Web
9  part2-1-sig1     Retrieve and plot EXFOR/ENDF cross sections
10 part2-2-da1an    EXFOR/ENDF angular distributions (Ei:fixed)
11 part2-3-da1ei    EXFOR/ENDF angular distributions (angle:fixed)
12 part2-4-de1eo    EXFOR/ENDF energy spectra of outgoing particles
13 part2-5-dae1eo   EXFOR/ENDF double differential cross sections
14 part2-6-fy       EXFOR/ENDF fission yeild (mass distribution)
15 part2-7-fye      EXFOR/ENDF fission yeild (energy distribution)
  part3:: EXFOR data renormalizations, corrections and recalculations
16 part3-1-auto1    Automatically renormalize EXFOR CS using old monitor CS and new standards
17 part3-2-user1    Correct data from EXFOR by user's program
18 part3-3-expert1  Apply experts corrections stored in X4Pro database
19 part3-4-auto2    Automatically renormalize EXFOR CS using old monitor CS and new standards, old and new decay data
20 part3-5-ratio2sig Retrieve ratios, calculate CS and uncertainties using latest standards
21 part3-6-daleg2sig Retrieve from EXFOR: Legendre coefficient L[0], calculate cross sections and errors
22 part3-7-legrs2da EXFOR LEG/RS and SIG from the same Entry; calc. missing Legendre coefficient L[0]; calc. angular d
istributions
23 part3-8-leg2r33  Retrieve Legendre coefficient calculate angular distributions store R33dat (draft of R33)
  part5:: Find mistakes in EXFOR library //EXFOR compilers
24 part5-0-sql      SQL to find zero values of PARITY, ERR-T and DATA-ERR
  part6:: Ratios to CS, Renormalization, Flags and scores //evaluators
25 part6-0-eval1flags Generate summary tables with evaluators' flags and scores
26 part6-1-eval2sig Retrieve CS, ratios, monit-data; renormalize to new standards and decay data; convert ratios to CS
```

x4pro> _

4) [1-5] - Run scripts runme.sh from demo-folders #1 to #5

```

C:\WINDOWS\system32\cmd.exe - bash x4pro.sh
x4pro> 1-5
--nn0=[1] --nn1:[5]
--Start example #1 Retrieve and plot EXFOR cross sections (modular Python-program)
---run: part1-1-sig/runme.sh
+ python -B sig0x.py
+ python -B sig0x2.py
+ set +x
--Finish example #1 21sec
--Start example #2 EXFOR angular distributions
---run: part1-2-da/runme.sh
+ python -B da0an.py show=0
+ set +x
--Finish example #2 4sec
--Start example #3 EXFOR partial angular distributions
---run: part1-3-dap/runme.sh
+ python -B dap0ei.py
+ set +x
--Finish example #3 4sec
--Start example #4 EXFOR energy spectra of outgoing particles
---run: part1-4-de/runme.sh
+ python -B de0eo.py
+ set +x
--Finish example #4 4sec
--Start example #5 EXFOR double differential cross sections
---run: part1-5-dae/runme.sh
+ python -B dae0eo.py
+ set +x
--Finish example #5 5sec
x4pro>

```

5) [C] - Clean temporary files. [Q] - Quit

```

C:\WINDOWS\system32\cmd.exe
x4pro> c
---Clean dirs: remove generated output files.
---Are you sure (y/n): y
--Start cleaning
---run: ./clean.sh
--clean-dir:3 part1-1-sig 8 files
--clean-dir:4 part1-2-da 4 files
--clean-dir:5 part1-3-dap 4 files
--clean-dir:6 part1-4-de 4 files
--clean-dir:7 part1-5-dae 4 files
--clean-dir:30 files:24
--Finish cleaning 6sec
x4pro> q
---EXIT---

G:\projects\zerkin\pdev\x4pro7-dist\x4pro14>

```